

Engineering Mathematics: Solving Systems of Linear Equations

Conceptual Overview: Augmented Matrices and Row Echelon Form (REF)

To solve a system of linear equations, we translate the system into a single mathematical object: the **Augmented Matrix**. This allows us to perform operations on the coefficients and constants simultaneously without the distraction of variable names.

The Logic of Row Operations

We use **Elementary Row Operations (EROs)** to transform the matrix into **Row Echelon Form (REF)**. The goal is to create a "staircase" pattern of zeros in the lower-left corner.

- Swapping:** Interchanging two rows.
- Scaling:** Multiplying a row by a non-zero constant.
- Pivoting:** Adding a multiple of one row to another to create zeros below a "pivot" element.

Once in REF, we use **Back-Substitution** to determine the values of the variables.

Step-by-Step Solution

Given System of Equations:

- $x + 2y - z = 3$
- $3x - y + 2z = 1$
- $2x - 2y + 3z = 2$
- $x - y + z = -1$

Step 1: Represent as an Augmented Matrix

We represent the system as $[A|B]$ where A is the coefficient matrix and B is the constant vector.

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 3 & -1 & 2 & 1 \\ 2 & -2 & 3 & 2 \\ 1 & -1 & 1 & -1 \end{array} \right]$$

Step 2: Eliminate x from Rows 2, 3, and 4

We use R_1 as the pivot row. The operations are:

- $R_2 \rightarrow R_2 - 3R_1$
- $R_3 \rightarrow R_3 - 2R_1$

- $R_4 \rightarrow R_4 - R_1$

$$\begin{bmatrix} 1 & 2 & -1 & 3 \\ 3 & -1 & 2 & 1 \\ 2 & -2 & 3 & 2 \\ 1 & -1 & 1 & -1 \end{bmatrix} (R_2 - 3R_1, R_3 - 2R_1, R_4 - R_1) \longrightarrow \begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & -7 & 5 & -8 \\ 0 & -6 & 5 & -4 \\ 0 & -3 & 2 & -4 \end{bmatrix}$$

Step 3: Simplify Row 4 (Optional but helpful for calculation)

To make calculations easier, we can swap R_2 and R_4 to use the -3 as a more manageable pivot.

$$\begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & -7 & 5 & -8 \\ 0 & -6 & 5 & -4 \\ 0 & -3 & 2 & -4 \end{bmatrix} (R_2 \leftrightarrow R_4) \longrightarrow \begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & -3 & 2 & -4 \\ 0 & -6 & 5 & -4 \\ 0 & -7 & 5 & -8 \end{bmatrix}$$

Step 4: Eliminate y from Rows 3 and 4

Using R_2 as the new pivot row:

- $R_3 \rightarrow R_3 - 2R_2$
- $R_4 \rightarrow R_4 - \frac{7}{3}R_2$ (Alternatively, use $3R_4 - 7R_2$ to avoid fractions)

Let's use $3R_4 - 7R_2 \rightarrow R_4$:

$$\begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & -3 & 2 & -4 \\ 0 & -6 & 5 & -4 \\ 0 & -7 & 5 & -8 \end{bmatrix} (R_3 - 2R_2, 3R_4 - 7R_2) \longrightarrow \begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & -3 & 2 & -4 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 1 & 4 \end{bmatrix}$$

Step 5: Eliminate z from Row 4

- $R_4 \rightarrow R_4 - R_3$

$$\begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & -3 & 2 & -4 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 1 & 4 \end{bmatrix} (R_4 - R_3) \longrightarrow \begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & -3 & 2 & -4 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Step 6: Back-Substitution Logic

The matrix is now in **Row Echelon Form**. We translate the rows back into equations:

1. From R_3 : $1z = 4 \implies z = 4$
2. From R_2 : $-3y + 2z = -4$
 - Substitute $z = 4$:
 - $-3y + 2(4) = -4$
 - $-3y + 8 = -4$
 - $-3y = -12 \implies y = 4$

3. From R_1 : $x + 2y - z = 3$

- Substitute $y = 4$ and $z = 4$:
- $x + 2(4) - (4) = 3$
- $x + 8 - 4 = 3$
- $x + 4 = 3 \implies x = -1$

Final Result

The system is consistent and has a unique solution:

- $x = -1$
- $y = 4$
- $z = 4$

Note on Consistency: Since the last row became all zeros ($0 = 0$), the fourth equation was redundant. Since the number of non-zero rows ($r = 3$) equals the number of variables ($n = 3$), the system has a unique solution.